MACHINE LEARNING METHODOLOGIES V1

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Understanding Feature Selection

In a machine learning dataset, **features** are the input variables used by a model to learn patterns and make predictions. In a typical DataFrame, these features are represented by columns, while the **target** is the variable we want the model to predict.

Feature selection is the process of deciding which of the available features should be included in a machine learning model.

A dataset may contain many columns, but not every column will contribute useful information. Some may be irrelevant, redundant, or contain very little variation. Feature selection helps us narrow the dataset down to variables that are more useful for the prediction task.

Why Feature Selection Matters

There are several reasons for reducing the number of features used by a model.

Reducing Unnecessary Complexity

When two models produce similar results, a model that achieves that performance with fewer features is generally easier to understand and work with.

Fewer inputs can make the model simpler and its results easier to interpret.

Improving Training Efficiency

A model has less information to process when unnecessary features are removed.

This can lead to:

- Shorter training times
- Lower computational requirements
- Reduced memory usage

This becomes especially valuable when dealing with datasets containing a large number of columns.

Supporting Model Performance

Some features may contain noise or information that does not help the model predict the target.

Removing irrelevant features can sometimes allow the model to focus more effectively on useful patterns and improve its performance on unseen data.

However, feature selection does not guarantee better performance in every situation.

When to Perform Feature Selection

Feature selection is generally carried out **before model selection and hyperparameter tuning**.

The reason is that the set of features being used can influence which model and hyperparameters perform well.

A simplified workflow can therefore be viewed as:

Data Preparation → Feature Selection → Model Selection → Hyperparameter Tuning → Evaluation

Care must also be taken to avoid **data leakage**. If feature selection uses information from the target, the selection process should be learned from the training data rather than allowing the test data to influence it.

Feature Selection Techniques

The resource covered four main approaches:

- Variance Threshold
- K-Best Features
- Recursive Feature Elimination (RFE)
- Boruta

Each method evaluates features from a different perspective.

Variance Threshold

Variance describes the degree to which values within a feature differ from one another.

If a feature has **zero variance**, every observation has the same value.

For example, imagine a column where every record contains the same country. Since there is no variation in that column, it provides no information for distinguishing one observation from another within that dataset.

The **Variance Threshold** technique removes features whose variance falls below a chosen threshold.

A common starting point is to remove features with zero variance.

Key idea

If a feature barely changes across observations, it may not provide enough information to be useful.

However, low variance does not automatically mean a feature has no predictive value. The threshold therefore needs to be chosen with the problem and data in mind.

K-Best Features

K-Best is a **filter-based feature selection method**.

It evaluates features using a statistical scoring method and then keeps the highest-scoring **K features**.

For example, if a dataset has many potential predictors and you decide to keep 10, K-Best ranks the features according to the selected scoring criterion and retains the top 10.

The important point is that the model itself does not need to be trained repeatedly to perform the initial selection.

Selecting an Appropriate Statistical Measure

The scoring technique used with K-Best depends on the types of the feature and target.

Categorical Feature and Categorical Target

The **Chi-square test** can be used to examine associations between categorical variables.

Categorical Feature and Numerical Target

A **t-test** can be appropriate in situations involving comparisons between two groups.

Numerical Feature and Categorical Target

A **t-test** can also be relevant when comparing a numerical variable across two categorical groups, provided its assumptions are appropriate.

Numerical Feature and Numerical Target

For numerical variables, measures such as:

- Pearson correlation
- Spearman rank correlation

can be used to assess relationships.

Pearson is mainly concerned with **linear relationships**, while Spearman works with the **rank ordering** of observations and can identify monotonic relationships that may not be linear.

The important lesson is that the scoring method must be appropriate for the type and characteristics of the data.

Recursive Feature Elimination (RFE)

Recursive Feature Elimination, commonly called **RFE**, takes a different approach.

It is a **wrapper method**, meaning that it uses a machine learning model as part of the feature-selection process.

The model evaluates the importance of the available features, after which the less important ones are removed. The model is then fitted again and the process continues until the desired number of features remains.

The process can be summarized as:

Train model → Determine feature importance → Remove weaker features → Train again → Repeat

Model Requirement

RFE needs an estimator capable of providing some measure of feature importance or coefficients.

Tree-based models are common examples because they can provide feature-importance information.

Main Advantage

Because the selection process is connected to a model, RFE can identify features that are useful in the context of that particular model.

Main Limitation

Repeatedly training a model can make RFE more computationally expensive than simpler filter-based approaches.

Boruta

Boruta is a model-based feature-selection technique designed to identify features that are genuinely relevant to the prediction task.

One of its main ideas is to avoid relying entirely on a manually chosen importance cutoff. Instead, Boruta creates a reference point using randomized versions of the original features.

This is why it can be described as reducing the need for subjective human decisions when determining which importance level should be considered sufficient.

How the Boruta Process Works

Creating Shadow Features

Boruta creates **shadow features** by taking the original features and randomly shuffling their values.

The purpose is to destroy the meaningful relationship between those features and the target while retaining similar general characteristics.

Measuring Importance

The original features and their shadow counterparts are passed through the underlying model, and feature importance is calculated.

Establishing the Benchmark

Boruta identifies the most important shadow feature.

Its importance becomes the benchmark against which the original features are compared.

Evaluating the Original Features

Each genuine feature is then compared with the shadow-feature benchmark.

If a real feature performs better than the benchmark, there is evidence that it contains useful information.

Features that fail to demonstrate sufficient importance can be rejected.

The Main Idea

Boruta essentially asks:

> **Does this actual feature provide more useful information than a randomized version of a feature?**

How the Methods Differ

Although all four techniques aim to reduce unnecessary features, they do not make their decisions in the same way.

Variance Threshold focuses on the amount of variation within a feature.

K-Best uses statistical relationships between features and the target to rank and select a specified number of features.

RFE relies on a machine learning model to repeatedly identify and eliminate less important features.

Boruta compares genuine features with randomized shadow features to determine whether their importance is meaningfully greater than random information.

An Easy Way to Think About the Methods

You can remember the four techniques through the questions they ask:

Variance Threshold:
"Does this feature contain enough variation?"

K-Best:
"Which features have the strongest statistical relationship with the target?"

RFE:
"Which features does this model consider important?"

Boruta:
"Is this feature more informative than a randomized feature?"

Feature Selection vs Feature Engineering

It is important not to confuse these two concepts.

Feature selection involves choosing which existing features should remain in the dataset.

Feature engineering involves creating or transforming features to produce potentially more useful inputs.

For example, selecting `Age` and `Income` from an existing dataset is feature selection. Creating a new feature such as `Income_per_Age` would be an example of feature engineering.

Both can play important roles in developing effective machine learning models.

Important Points to Remember

Feature selection does not mean simply reducing the number of columns as much as possible. The objective is to retain the variables that provide useful information while removing those that are unnecessary.

The usefulness of a feature can also depend on the model being used. A feature that contributes to one model may not have the same importance in another.

Most importantly, feature selection must be performed carefully to avoid **data leakage**. Information from the test set should not influence the process used to decide which features the model will learn from.

Final Takeaway

Feature selection is an important part of machine learning because **more features do not automatically produce a better model**.

The goal is to build a useful set of predictors while reducing irrelevant information, unnecessary complexity, and computational cost.

The four approaches covered can be remembered as:

Variance Threshold → Remove features with insufficient variation

K-Best → Select the top K features using statistical scores

RFE → Repeatedly eliminate features based on model importance

Boruta → Compare real features against randomized shadow features

Ultimately, good feature selection is about giving the model **the right information rather than simply giving it more information**.